

Arya Mohammadi

U.S. Citizen | 424-204-2995 | aryamohammadi.dev@gmail.com
linkedin.com/in/aryamshahi | github.com/aryamohammadi | aryamohammadi.github.io

EDUCATION

University of California - San Diego

Expected June 2028

M.S. in Computer Science

B.S. in Computer Science and Mathematics

GPA: 3.70/4.00

Coursework: Advanced Data Structures, Design & Analysis of Algorithms, Software Engineering, Systems Programming, Artificial Intelligence, Data Science Principles, Graph Theory, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Tools: Next.js, React, Node.js, Express, FastAPI, OpenSearch, MongoDB, SQLite, Docker, Git, Linux, Tailwind CSS, Leaflet, bcrypt, Twilio, Vercel, Jest, Supertest, GoogleTest, CMake, gdb, Valgrind

Concepts: REST APIs, CI/CD, Unit & E2E Testing, Authentication (JWT, HMAC), RAG & Hybrid Retrieval, Model Context Protocol (MCP)

AI Development: Cursor, AI-assisted code generation, refactoring, testing, debugging, and documentation; validation through automated tests, code review, and security checks

EXPERIENCE

IBM — watsonx Orchestrate

Jun 2026 – Sep 2026

Software Engineer Intern

Austin, TX

- Built the team's first search-accuracy test suite using Python and OpenSearch uncovering that AI-powered semantic search had been silently broken since launch across 2 IBM products serving 5 AI agents
- Designed the fix using OpenSearch search pipelines and feature-flagged updates to 2 Python backend services, enabling a safe, one-setting rollout with zero deployment risk; verified with 15 unit tests in pytest
- Fixed a ranking bug in the Python search code that let long documents crowd out the single best answer by up to ~7x, and restored 2 search features that had never worked; merged into production with a 5-test regression suite
- Audited an AI platform of 13 agents maintained by ~20 engineers using Python and Node.js code review, documenting 12 verified bugs (3 high severity), including a security flaw that could leak access credentials to an attacker
- Tested 30+ REST API endpoints against a live QA cluster (assigned 1 of 4 sections, completed all 4), filing 30+ defect reports with suggested fixes, including server crashes on valid requests

Arya Electric LLC | aryaelectric.com | [engineering write-up](#)

Jun 2025 - Jun 2026

Software Engineer & Co-Founder

Irvine, CA

- Drove \$30k+ in booked revenue for a licensed electrical contractor by building a Next.js/React lead-generation platform with 15 serverless APIs, Twilio SMS + email quote intake, and a Leaflet map of completed jobs across 67 cities.
- Replaced the CMS with an authenticated owner portal for batch photo uploads; browser-side compression cuts 10 MB images to 200–500 KB, while cache revalidation publishes updates with zero redeloys.
- Eliminated silent data loss in batch uploads by reproducing a CDN stale-read race and redesigning persistence around immutable, versioned index files.
- Extended the platform to a second production contractor site for NOOR Electric Inc. by reusing the quote-intake and notification infrastructure.

University of California, Riverside

Sep 2024 - Jun 2025

Computer Science and Mathematics Tutor

Riverside, CA

- Tutored 400+ students over 600+ hours in data structures, algorithms, and C++ systems programming.
- Debugged 100+ student C++ codebases one-on-one; authored 50+ practice problems adopted by 3 other tutors.

PROJECTS

AgentSight — Cursor IDE Support (Open Source) | *Rust, SQLite, Python* | [PR #149 \(merged\)](#) | [GitHub](#)

- Shipped AgentSight's first IDE integration by adding Cursor support to the 566-star open-source AI-agent observability platform; merged into v1.0.4 and released on crates.io and npm.
- Reverse-engineered Cursor's undocumented session storage (JSONL transcripts joined to a 500 MB SQLite state DB) across macOS, Linux, and Windows; validated the schema on 496 records from two independent installs.
- Built zero-config session discovery, parsing, and enrichment in Rust, handling subagent delegation, torn writes, and WAL-locked databases; verified against live installs and exactly matched token totals for a 15.6M-token session.

DormDuos | *React, Express, MongoDB, JWT* | ucr-housing-frontend.vercel.app | github.com/aryamohammadi/Dormduos

- Founded a 500+ member UCR housing community and built its integrated marketplace with JWT authentication, bcrypt password hashing, and input sanitization against NoSQL injection.
- Optimized MongoDB with compound indexes; benchmarked 28k req/sec on 10k+ listings with 100% branch coverage.

Battlestar RPG | *C++, STL, GoogleTest, CMake* | github.com/aryamohammadi/Battlestar-RPG

- Built a terminal RPG engine in C++ with a 5-person team, implementing a MaxHeap combat scheduler with $O(\log n)$ turn updates, inventory and save/load persistence, and 2.3k lines of GoogleTest coverage across core game systems.